

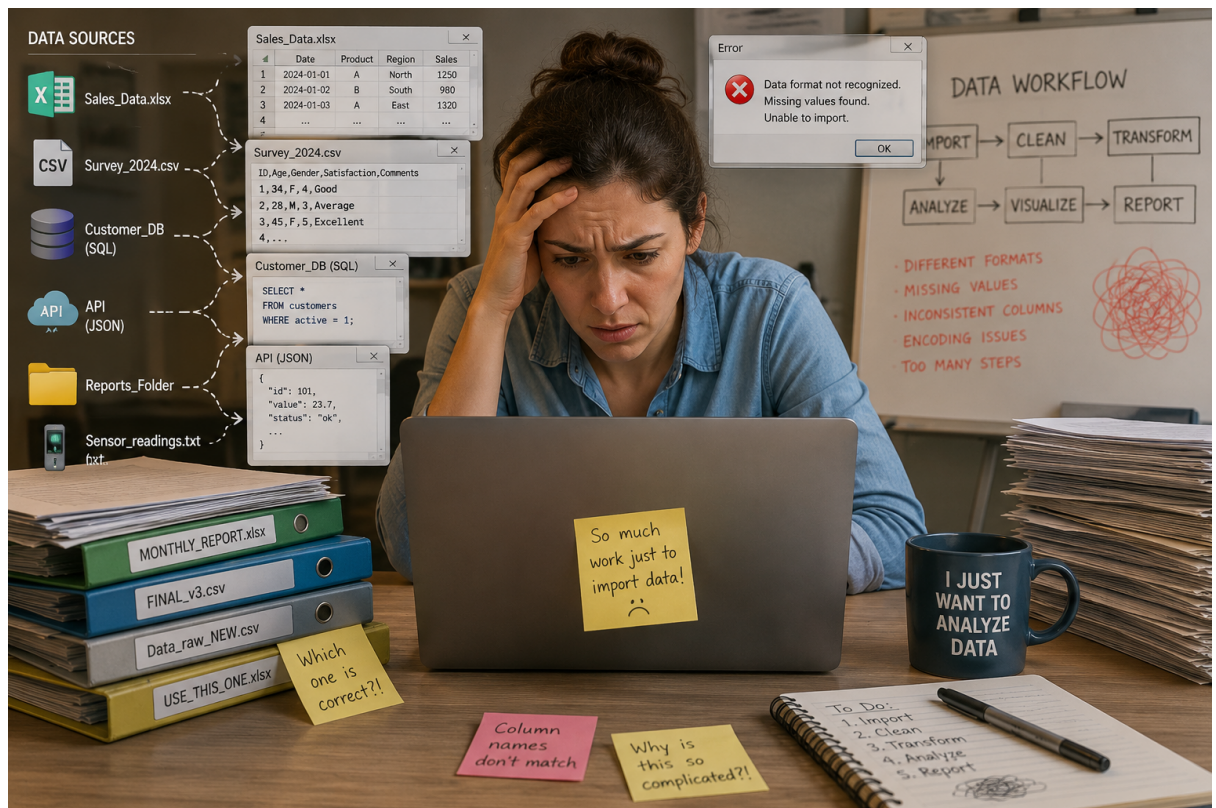
Importing Data in R: Exploring Various Packages and Their Advantages: Data Analysis for Decision-Making (SS 2026)

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Abstract

This document describes the importance of importing data in R, introducing the packages and their advantages.

1 Why Do We Import Data?



2 Why Do We Import Data? Why with R?

- Accessing the “Raw” Reality
- The Reproducibility Imperative
- Auditable
- Efficiency and Automation
- Handling Complexity

3 Different Data Types

1. Rectangular/Tabular Data
2. Proprietary Spreadsheets
3. Statistical Software Files
4. Relational Databases
5. Spatial Data
6. Base Text and Fixed-Width Files
7. Scientific/Specialized Data

4 Graphical and Programmatic Implementation by Data Type

4.1 Standard Tabular Data (.csv, .tsv)

- **Graphical:** In R studio click on the File and then Import dataset and then you can see two options for importing text file, as shown in the handout.
- **Code:**

From text Base

```
df_base <- read.csv("data.csv")
df_base
```

From text readr

```
library(readr)
df_readr <- read_csv("data.csv")
df_readr
```

4.2 Proprietary Excel Files (.xls, .xlsx)

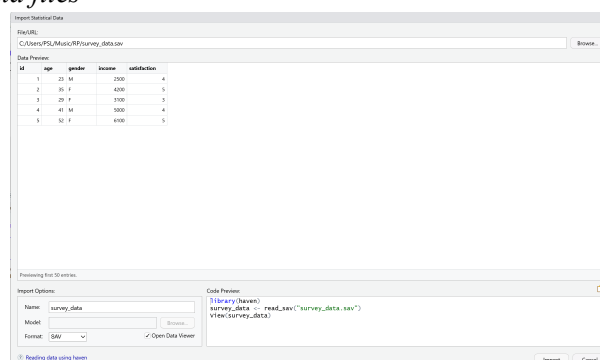
install Tideyverse

- **Graphical:** Select “From Excel” in the RStudio wizard. Then you you can select from which sheet you want to import data, and also define a cell range (e.g., “A1:E20”) graphically.
- **Code** Chapman University (2026):

```
library(readxl)
df_excel <- read_excel("file.xlsx")
df_excel
```

4.3 Statistical Software Files (.sav, .sas7bdat, .dta)

- **Graphical:** Select “From SPSS/SAS/Stata” in RStudio.
- **Code** R Core Team (2015):

Figure 1*Importing from .sav or .dta files*

```
library(haven)
# Import SPSS file
df_spss <- read_sav("survey_data.sav")
# Import Stata file
df_stata <- read_dta("economic_data.dta")
# Preview imported data
head(df_spss)
head(df_stata)
```

4.4 Spatial / GIS Data (.shp, .tif, .kml)

- **Code Bivand et al. (2008) :**

```
library(sf)
# Read shapefile
spatial_data <- st_read("sample_shapefile.shp")
# Preview spatial object
print(spatial_data)
```

4.5 Base Text and Fixed-Width Files (.txt, .dat, .fwf)

- **Graphical:** Use the “From Text (base)” option in the RStudio wizard.
- **Code R Core Team (2015):**

```
# Import fixed-width file
df_fwf <- read.fwf(
  "legacy_data.fwf",
  widths = c(3, 6, 5),
  col.names = c("ID", "Name", "Score")
)

df_fwf
```

4.6 Scientific Spectral Data (.jdx, .abs, .cnv, .nc)

- **Graphical:** Utilize specialized package GUIs or high-level functions that automate folder-wide imports Gruson et al. (2019).

- **Code Gruson et al. (2019) :**

```
library(lightr)

# Import spectral data
spec_data <- lr_get_spec("spectrum.jdx")

# Show spectral data
head(spec_data)
```

4.7 Cloud Sheets (Google Sheets/ URLs)

- **Code: Chapman University (2026)**

```
library(google Sheets4)
# Import Google Sheet
gs4_deauth()
sheet_data <- read_sheet(
  "https://docs.google.com/spreadsheets/d/1BoD9bXwbxm1NJWhEnk6BVXUxuxWbWArQWVT3LIk2PQ/"
)

sheet_data
```

4.8 Hierarchical Data (.json)

- **Graphical:** While RStudio does not have a dedicated “JSON Wizard” in the same way it does for Excel or CSV, users can often preview JSON structures through the File Pane or by using specialized viewing packages.
- **Code R Core Team (2015) :**

```
library(jsonlite)
df_json <- fromJSON("data.json")
df_json
```

5 The Comparison Between Packages

- **Tabular Data:** Very fast import; creates tibbles / Limited by RAM.
- **Excel Sheets:** No Java required; supports multiple sheets / Limited scalability.
- **Statistical Files:** Preserves labels and metadata / Limited newest-version support.
- **Databases (SQL):** Handles large datasets / Requires SQL knowledge.
- **Spatial / GIS:** Supports CRS metadata / Complex for beginners.
- **Hierarchical (Web):** Standard for APIs and JSON / Nested data can be complex.
- **Batch Import:** Automates multi-file import / Requires functional programming.
- **Cloud Services:** Direct cloud access / Requires OAuth authentication.

6 Conclusion

Mastering data import is about matching the right package to the right data structure while respecting hardware limits. Choosing programmatic scripts ensures that the decision-making process is transparent, scalable, and reproducible Bivand et al. (2008), Huber (2025).

7 Class Assignment

Download the files from here [link](#).

7.1 Help The HR Manager!

1. You are helping a HR manager in the company. She has given you CSV file about the personnel information (off-course after signing relative Non-disclosure agreement!). This data file is changing every month and it is a real pain for her to create the report every time! She knows you have R knowledge! and ask you to prepare a dynamic file by import the data into R, so she can easily prepare report very quickly in the beginning of every month. go ahead and do first step which is importing the [exe1.csv](#) file into R Studio.

7.2 Everyday Sales Report!

2. You are a manager in a company and the sale department gives you sales reports in excel file every morning. consider that you have recieved [exe2.xlsx](#) file today. Import it to R Studio.

7.3 Spare Parts Sheet

3. Imagine that you and your colleagues in the company are working on a google spreadsheet, together. now you want to import data from it into R. Use this [Link](#) and import it in the R Studio. if you had problem with the link to Google spreadsheet, upload it (it is available in the exercise zip folder) in your own account, and then use your own link.

8 References

- Bivand, R. S., Pebesma, E. J., & Gómez-Rubio, V. (2008). *Applied spatial data analysis with r*. Springer Science & Business Media. <https://doi.org/10.1007/978-0-387-78171-6>
- Chapman University. (2026). *R programming: Importing data files into r*. <https://libguides.chapman.edu/R/import>
- Gruson, H., White, T., & Maia, R. (2019). Lightr: Import spectral data and metadata in r. *Journal of Open Source Software*, 4(43), 1857. <https://doi.org/10.21105/joss.01857>
- Huber, S. (2025). *How to use r for data science*. <https://hubchev.github.io/ds/>
- R Core Team. (2015). *R data import/export*. R Foundation for Statistical Computing. <https://cran.r-project.org/doc/manuals/r-release/R-data.pdf>

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